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EXP 1: FOOTPRINTING ON THE TARGET NETWORK USING SEARCH ENGINES, WEB SERVICES, AND SOCIAL NETWORKING SITES

AIM:

To perform passive footprinting on the authorized target domain rajalakshmi.org using search engines, public web services, and social networking sites to collect publicly available information about the organization, website technologies, domain details, and subdomains.

THEORY:

Footprinting (also called reconnaissance) is the very first step of ethical hacking. It means collecting as much information as possible about a target before attacking it. This is a passive activity — we are not sending any traffic to the target's servers, we are only looking at information that is already public.

Information collected in this step can include:

- Company name, employee names, job titles
- Email address formats used by the company
- Technologies used by the company (from job postings)
- Physical location, phone numbers
- Social media activity that may reveal internal information

There are two types of footprinting:

1. Passive Footprinting – Collecting information without directly contacting the target (search engines, social media).
2. Active Footprinting – Directly interacting with the target's systems (covered in later experiments).

This experiment focuses only on passive footprinting.

TOOLS REQUIRED:

- A web browser (Firefox, on Kali Linux)
- Google Search (with advanced search operators, also called "Google Dorks")
- Shodan (search engine for internet-connected devices) – https://www.shodan.io
- theHarvester (built into Kali Linux)

- LinkedIn, Twitter/X, Facebook (for social media footprinting)

PROCEDURE:

Step 1: Google Dorking (Advanced Search Operators)

Open Firefox and use special Google search operators to find hidden or specific information about the target organization. Example searches (replace targetcompany.com with your lab's given target domain):

```
site:targetcompany.com
site:targetcompany.com filetype:pdf
site:targetcompany.com intitle:"index of"
site:targetcompany.com inurl:admin
"targetcompany.com" intext:"confidential"
```

Explanation of operators:

- site: restricts results to a specific domain
- filetype: finds documents of a specific type (pdf, docx, xlsx)
- intitle: finds pages with a specific word in the title
- inurl: finds pages with a specific word in the URL

Step 2: Use theHarvester to Collect Emails and Subdomains

Open a terminal in Kali Linux and run:

```
theHarvester -d targetcompany.com -b google
```

- -d specifies the target domain
- -b specifies the data source (google, bing, duckduckgo, etc.)

You can also search multiple sources at once:

```
theHarvester -d targetcompany.com -b all
```

This tool will return employee email addresses, subdomains, and sometimes IP addresses that are publicly listed.

Step 3: Search Shodan for Exposed Devices

Go to https://www.shodan.io in the browser and search:

```
org:"Target Company Name"
```

Shodan shows devices connected to the internet (routers, cameras, servers) that belong to the searched organization, along with open ports and running services.

Step 4: Social Media Footprinting

Search the target company name on LinkedIn, Twitter/X, and Facebook.

- On LinkedIn, look at the "People" section to find employee names and job titles.
- Job titles like "Network Administrator using Cisco ASA" can reveal what technology the company uses.
- Check if employees post pictures of their workstations (badges, screens, whiteboards can leak information).

Step 5: Record Your Findings

Make a small table of everything you found:

Information Type	Data Found	Source
Employee Names	...	LinkedIn
Email Format	firstname.lastname@company.com	theHarvester
Subdomains	mail.company.com	theHarvester
Exposed Devices	Router at IP x.x.x.x	Shodan

OUTPUT:

INFORMATION	RESULT
Target Domain	rajalakshmi.org
Indexed Pages	Home page, Regulations 2023, Blogs
PDF Files	Rajalakshmi School of Architecture, Chennai; Minutes of the 18th Academic Council Meeting; Minutes of the 11th Governing Body Meetings.
Technologies (Using BuiltWith)	Next.js; Google Analytics; Microsoft Clarity; Google Tag Manager
Subdomains (Using crt.sh)	www.rajalakshmi.org, architecture.rajalakshmi.org, admissions.rajalakshmi.org
Social Media	Linkedln: www.linkedin.com/school/rajalakshmi-engineering-college/ Instagram:www.instagram.com/rec_chennai?utm_source=ig_web_button_share_sheet&igsh=ZDNlZDc0MzIxNw==

RESULT:

Passive footprinting of rajalakshmi.org was successfully performed using public sources. Indexed pages, PDF files, subdomains, website technologies, and official social media information were identified and documented without active scanning or unauthorized access.